

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B.Tech (R&AI) (CBCS) Degree Examinations
 December 2023

RI2105-1 – DATA STRUCTURES AND ALGORITHMS

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five** full questions choosing **Two** full questions from **Unit – I & Unit – II** each and **One** full question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. **What is a binary tree?**
 - A) A tree with only one node
 - B) A tree with two child nodes for each parent node
 - C) A tree with no child nodes
 - D) A tree with more than two child nodes for each parent node
2. **In a binary search tree (BST), what is the property that ensures that the left subtree contains only values less than the parent node, and the right subtree contains values greater than the parent node?**
 - A) AVL property
 - B) Heap property
 - C) Binary property
 - D) BST property
3. **What is the primary goal of the "Big Omega" notation in algorithm analysis?**
 - A) To provide a lower bound on the time complexity of an algorithm
 - B) To describe the average-case behavior of an algorithm
 - C) To provide a formal proof of correctness for an algorithm
 - D) To provide an upper bound on the time complexity of an algorithm
4. **Which sorting algorithm is known for its stability, meaning that the relative order of equal elements remains unchanged after sorting?**
 - A) Quick Sort
 - B) Merge Sort
 - C) Bubble Sort
 - D) Selection Sort
5. **What is the primary advantage of a breadth-first search (BFS) over a depth-first search (DFS)?**
 - A) BFS has better memory usage.
 - B) BFS always finds the shortest path.
 - C) BFS is faster in most cases.
 - D) BFS is easier to implement.
6. **Which data structure is used to implement undo functionality in text editors?**
 - A) Stack
 - B) Queue
 - C) Linked List
 - D) Hash Table
7. **Which data structure is used to represent a hierarchical structure and is often used for representing file systems?**
 - A) Stack
 - B) Queue
 - C) Tree
 - D) Heap
8. **What is the primary goal of the "Big O" notation in algorithm analysis?**
 - A) To provide a lower bound on the time complexity of an algorithm
 - B) To describe the average-case behavior of an algorithm
 - C) To provide a formal proof of correctness for an algorithm
 - D) To provide an upper bound on the time complexity of an algorithm
9. **What does the term "asymptotic analysis" refer to in the context of algorithms?**
 - A) Analyzing the best-case behavior of an algorithm
 - B) Analyzing the worst-case behavior of an algorithm
 - C) Analyzing the average-case behavior of an algorithm
 - D) Analyzing the behavior of an algorithm as the input size approaches infinity
10. **In the context of searching algorithms, what is "binary search" often used for?**
 - A) Searching for files on a hard drive
 - B) Searching for a specific value in a sorted list
 - C) Searching for a pattern in a text document
 - D) Searching for the shortest path in a graph

11. Which algorithm is used to find the minimum spanning tree in a connected, undirected graph with weighted edges?
 - A) Dijkstra's Algorithm
 - B) Prim's Algorithm
 - C) Bellman-Ford Algorithm
 - D) Kruskal's Algorithm
12. In Big O notation, what does $O(1)$ represent?
 - A) Linear time complexity
 - B) Quadratic time complexity
 - C) Constant time complexity
 - D) Exponential time complexity
13. What is an algorithm?
 - A) A type of data structure
 - B) A step-by-step procedure to solve a problem
 - C) A mathematical formula
 - D) A programming language
14. Which data structure allows us to implement graphs most efficiently?
 - A) Linked List
 - B) Stack
 - C) Queue
 - D) Adjacency List
15. In a binary tree, which node is at the very top?
 - A) Left child
 - B) Right child
 - C) Root
 - D) Leaf node
16. What are the disadvantages of arrays?
 - A) Index value of an array can be negative
 - B) Elements are sequentially accessed
 - C) Data structure like queue or stack cannot be implemented
 - D) There are chances of wastage of memory space if elements inserted in an array are lesser than the allocated size
17. Which data structure is used for implementing recursion?
 - A) Stack
 - B) Queue
 - C) List
 - D) Array
18. Which data structure is needed to convert infix notation to postfix notation?
 - A) Tree
 - B) Branch
 - C) Stack
 - D) Queue
19. What is the value of the postfix expression $6\ 3\ 2\ 4\ +\ -\ *?$
 - A) 74
 - B) -18
 - C) 22
 - D) 40
20. The prefix form of $A-B/(C * D ^ E)$ is?
 - A) $-A/B^*C^*DE$
 - B) $-A/BC^*^*DE$
 - C) $-ABCD^*^*DE$
 - D) $-/*^*ACBDE$

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Classify the various types of data structures with the suitable example.	8	L*2	1	1
	b) Develop a C program to demonstrate the INSERT and DELETE operation of a QUEUE and DISPLAY the QUEUE contents.	8	L3	2	1
2.	a) Explain structure and pointers with the suitable example in programming.	8	L2	1	1
	b) Develop a C program to evaluate a postfix expression using STACK.	8	L3	2	1
3.	a) List and explain the fundamental operations that can be performed on data structures? With the suitable example.	8	L2	1	1
	b) Develop a STACK data structure using Singly Linked List.	8	L3	2	1
Unit – II					
4.	a) With the suitable example explain various representation of Trees.	8	L2	3	1
	b) Apply Selection sort to sort the following elements and write the algorithm for the same. 89, 45, 68, 90, 29, 34, 17	8	L3	4	1
5.	a) What is time complexity? Explain various asymptotic notations to find the time complexities of the algorithms.	8	L2	4	1
	b) Apply merge sort to sort the following elements and write the algorithm for the same. 9, 7, 1, 89, 56, 34, 32, 11, 40, 38.	8	L3	4	1

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|-------|---|---|----|---|---|
| 6. a) | What is binary search tree? With the suitable example explain traversal of binary search trees. | 8 | L2 | 3 | 1 |
| b) | Develop a program in C for the following operations on doubly Linked List (DLL). | | | | |
| i) | Insert front | | | | |
| ii) | Delete front | | | | |
| iii) | Exit | 8 | L3 | 2 | 1 |

Unit – III

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|-------|--|---|----|---|---|
| 7. a) | With the suitable example explain dynamic programming. | 8 | L2 | 5 | 1 |
| b) | Explain how greedy method is applied to find the minimum spanning tree for a weighted graph. | 8 | L3 | 5 | 1 |
| 8. a) | Explain BFS with suitable example. | 8 | L2 | 5 | 1 |
| b) | Apply backtracking to solve the N queens problem and explain. | 8 | L3 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
